

SpaceX Falcon 9 First-Stage Landing Prediction

- IBM Data Science Capstone
- Business question: Can launch characteristics help predict booster landing success?
- GitHub: [ADD YOUR GITHUB REPOSITORY URL]



Executive Summary

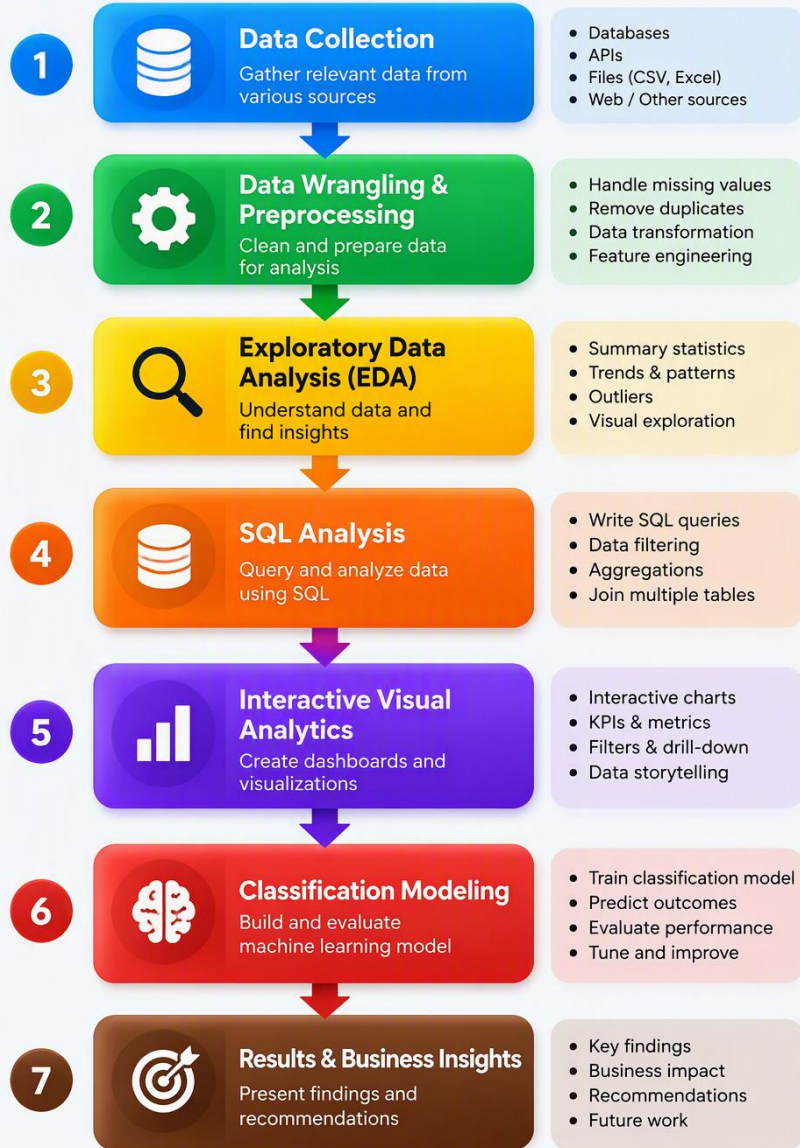
Outcome: a reproducible comparison of Logistic Regression, SVM, and Decision Tree.

Insert actual model metrics after running the notebook

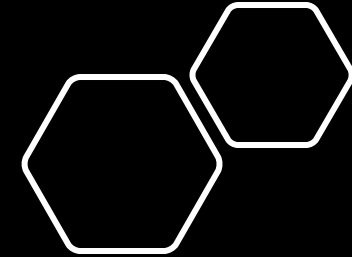


Data Analytics & Machine Learning Workflow

From Data to Decisions



Transforming Data into Meaningful Decisions



Introduction & Problem



Reusable first stages affect launch economics.



Goal: predict landing outcome from historical launch attributes.



Use case: support preliminary competitive bid and risk analysis—not replace engineering judgment

Data Collection — SpaceX API



Request structured launch records using HTTP GET.



Parse JSON responses into a tabular dataset.



Inspect response status, schema, missing fields, and duplicates.



GitHub: [ADD YOUR GITHUB REPOSITORY URL]

Data Collection — Web Scraping



Request public launch-history page.



Parse HTML tables with BeautifulSoup/pandas.



Normalize headers and reconcile scraped fields with API data.



Validate changes in page structure before rerunning.



GitHub: [ADD YOUR GITHUB REPOSITORY URL]

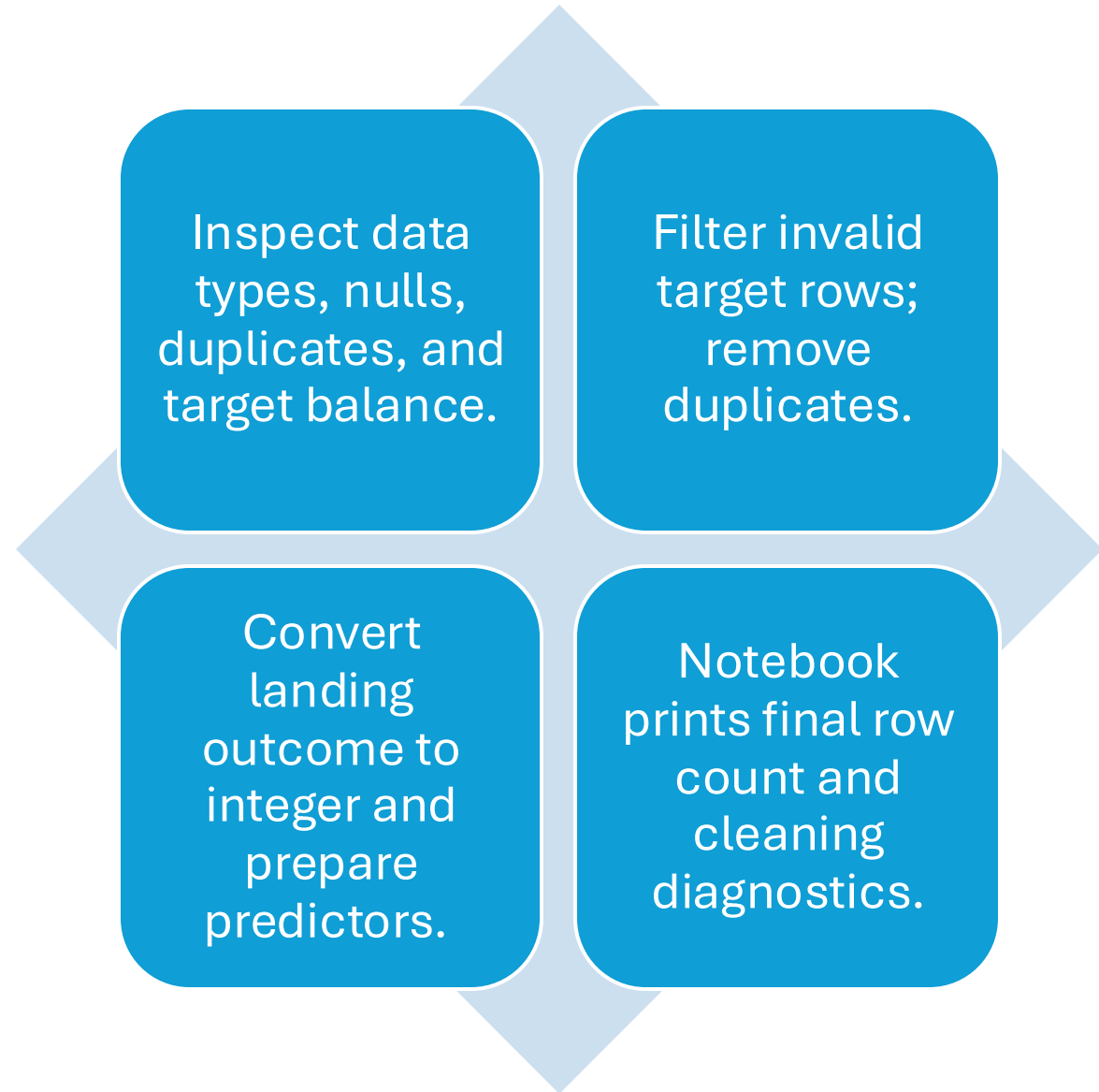
Data Collection — Web Scraping

Shape: (90, 18)

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 90 entries, 0 to 89
Data columns (total 18 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   FlightNumber        90 non-null    int64
1   Date                90 non-null    object
2   BoosterVersion      90 non-null    object
3   PayloadMass         90 non-null    float64
4   Orbit               90 non-null    object
5   LaunchSite          90 non-null    object
6   Outcome              90 non-null    object
7   Flights              90 non-null    int64
8   GridFins            90 non-null    bool
9   Reused               90 non-null    bool
10  Legs                 90 non-null    bool
11  LandingPad           64 non-null    object
12  Block                90 non-null    float64
13  ReusedCount          90 non-null    int64
14  Serial               90 non-null    object
15  Longitude             90 non-null    float64
16  Latitude              90 non-null    float64
17  Class                 90 non-null    int64
dtypes: bool(3), float64(4), int64(4), object(7)
memory usage: 10.9+ KB
None
```

Data Wrangling Methodology



Data Wrangling

...

	missing_values
LandingPad	26
FlightNumber	0
Date	0
BoosterVersion	0
Orbit	0
PayloadMass	0
Outcome	0
Flights	0
GridFins	0
LaunchSite	0
Reused	0
Legs	0
Block	0
ReusedCount	0
Serial	0

Duplicate rows: 0
Clean shape: (90, 18) | Target distribution:

	count
Successful	60
Unsuccessful	30

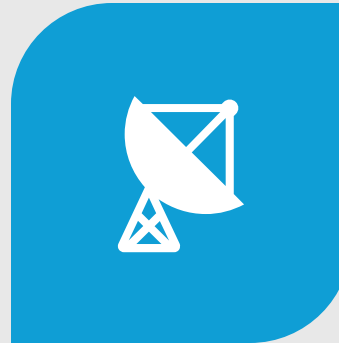
EDA with Data Visualization



OUTCOME COUNT PLOT:
ASSESS CLASS BALANCE.



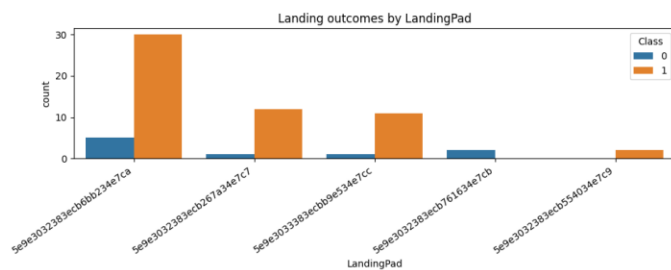
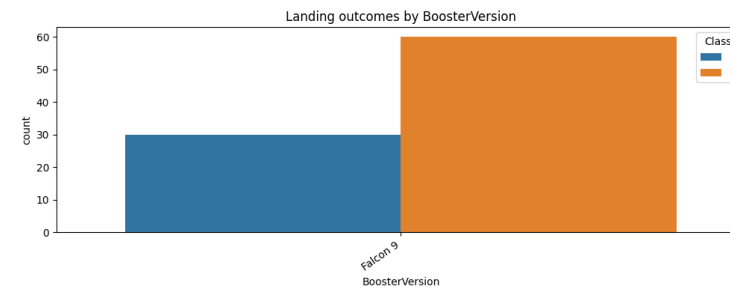
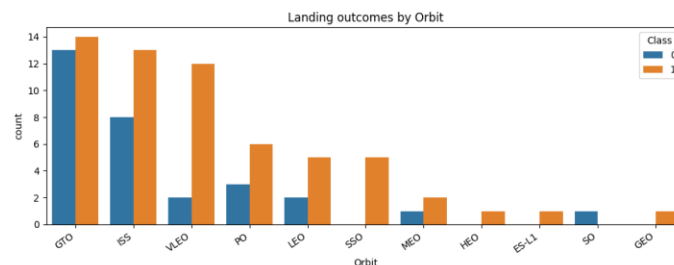
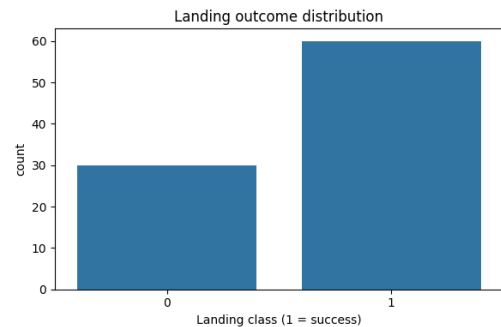
GROUPED BARS: COMPARE
OUTCOMES BY SITE, ORBIT,
BOOSTER, AND LANDING PAD.



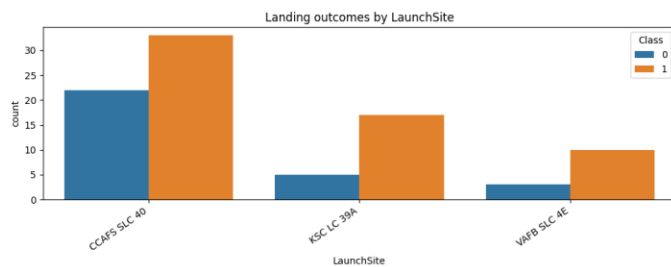
PAYLOAD PLOT: INSPECT
POSSIBLE RELATIONSHIP WITH
OUTCOME.



INTERPRET ASSOCIATION
CAUTIOUSLY; CORRELATION IS
NOT CAUSATION.



EDA with Data Visualization



EDA with SQL

Assess

Outcome count plot: assess class balance.

Compare

Grouped bars: compare outcomes by site, orbit, booster, and landing pad.

Inspect

Payload plot: inspect possible relationship with outcome.

Interpret

Interpret association cautiously; correlation is not causation.

EDA with SQL

launching_class	launches
0	1
1	0

Summary by LaunchSite

	category	launches	success_rate
0	KSC LC 39A	22	0.772727
1	VAFB SLC 4E	13	0.769231
2	CCAFS SLC 40	55	0.600000

Summary by Orbit

	category	launches	success_rate
0	SSO	5	1.000000
1	HEO	1	1.000000
2	GEO	1	1.000000
3	ES-L1	1	1.000000
4	VLEO	14	0.857143
5	LEO	7	0.714286
6	PO	9	0.666667
7	MEO	3	0.666667
8	ISS	21	0.619048
9	GTO	27	0.518519
10	SO	1	0.000000

Summary by LandingPad

	category	launches	success_rate
0	5e9e3032383ecb554034e7c9	2	1.000000
1	5e9e3032383ecb267a34e7c7	13	0.923077
2	5e9e3033383ecbb9e534e7cc	12	0.916667
3	5e9e3032383ecb6bb234e7ca	35	0.857143
4	None	26	0.192308
5	5e9e3032383ecb761634e7cb	2	0.000000

Interactive Visual Analytics

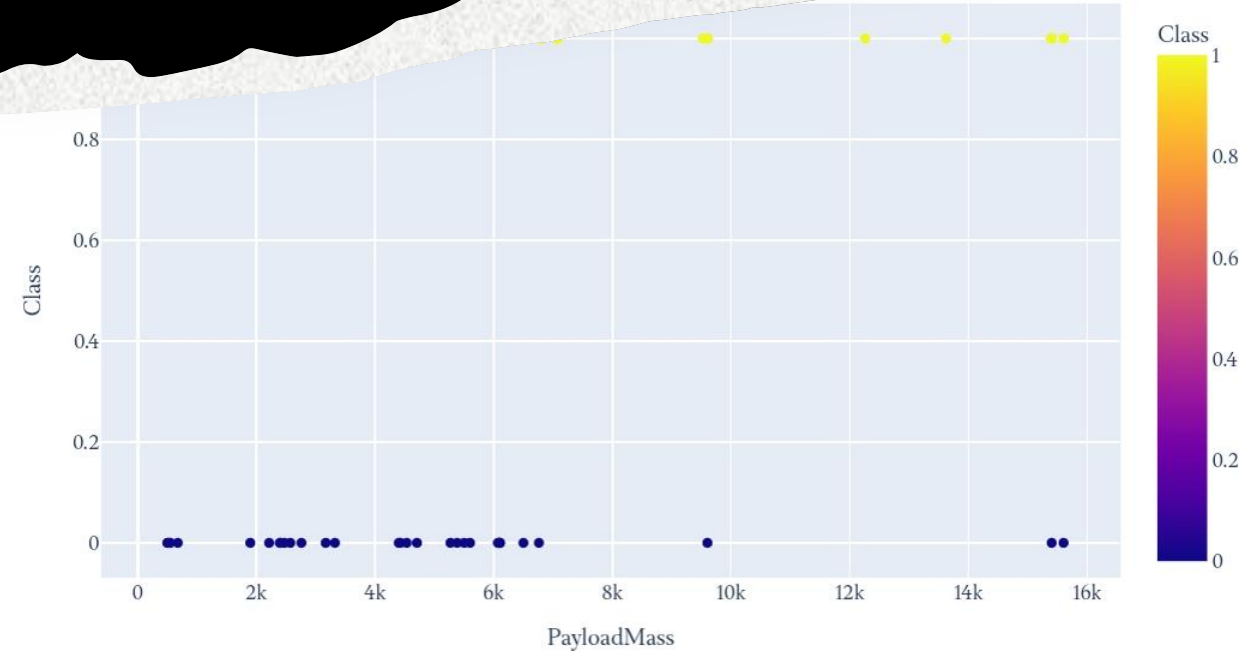
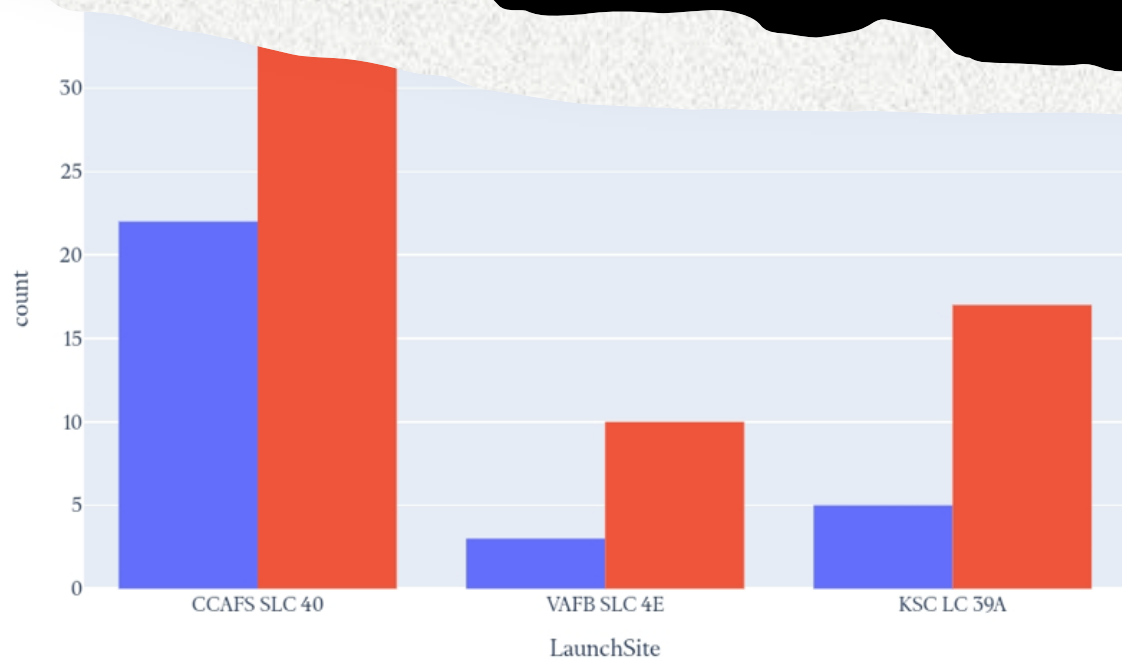
Plotly provides interactive categorical and payload views.

Dash can turn the same analysis into a filterable web app.

Document filters, chart purpose, and key patterns after execution.

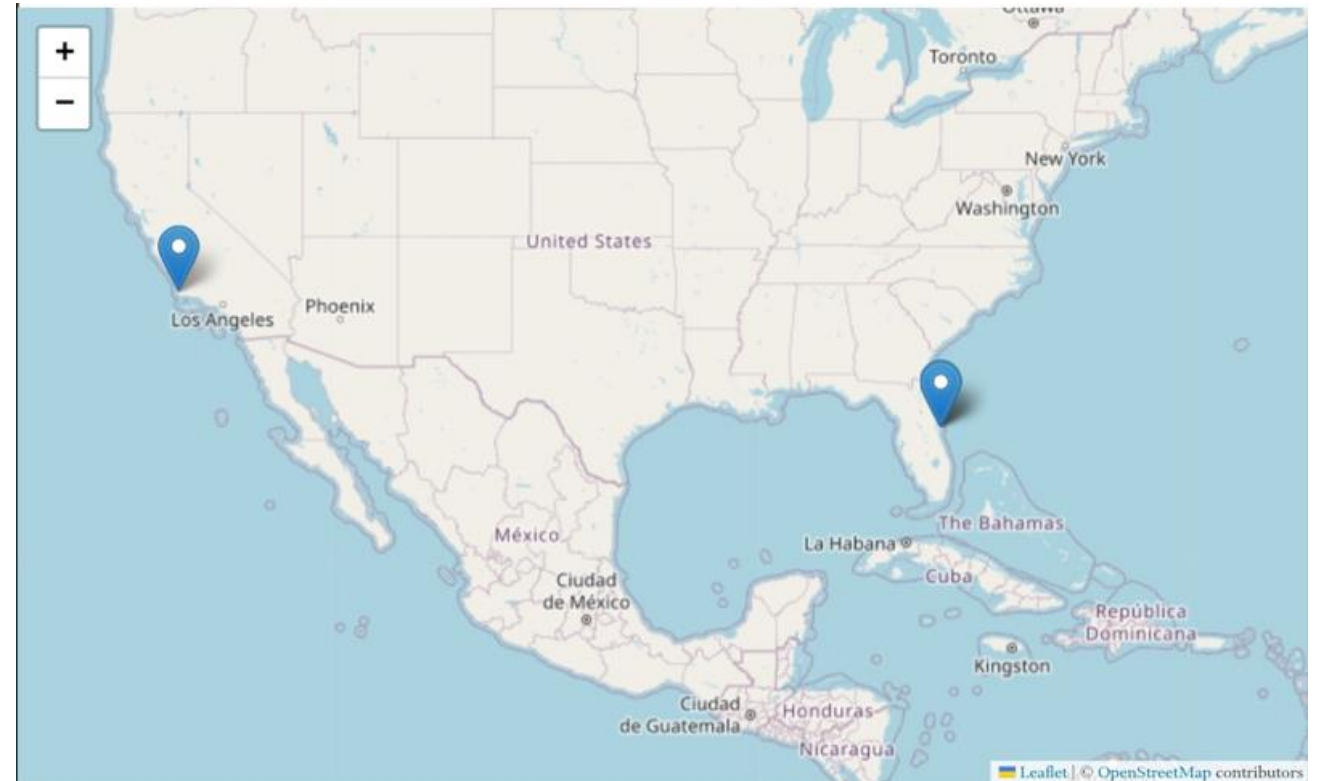
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InteractiveVisualAnalytics



Folium Launch-Site Map

- Map markers: CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E.
- Use the map to contextualize geography and site proximity.
- Proximity does not by itself explain landing success.
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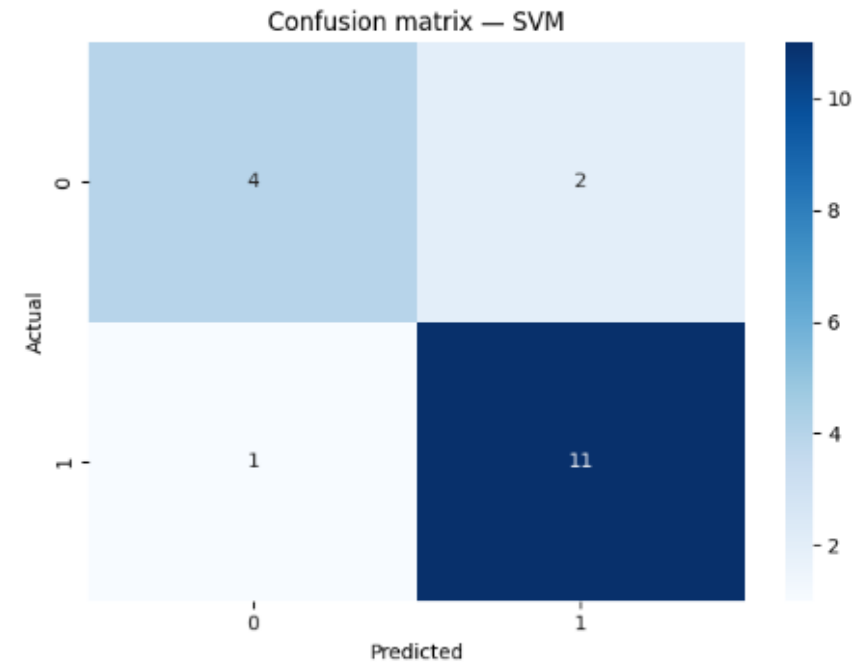
Predictive Analysis

- Train/test split: 80/20; stratify target where possible.
- Models: Logistic Regression, SVM, Decision Tree.
- Metrics: accuracy, precision, recall, F1; inspect confusion matrix.
- Select best model by test F1, then discuss limitations

	Model	Accuracy	Precision	Recall	F1
1	SVM	0.833	0.846	0.917	0.880
0	Logistic Regression	0.778	0.833	0.833	0.833
2	Decision Tree	0.722	0.769	0.833	0.800

Best model by test F1: SVM

	precision	recall	f1-score	support
0	0.80	0.67	0.73	6
1	0.85	0.92	0.88	12
accuracy			0.83	18
macro avg	0.82	0.79	0.80	18
weighted avg	0.83	0.83	0.83	18



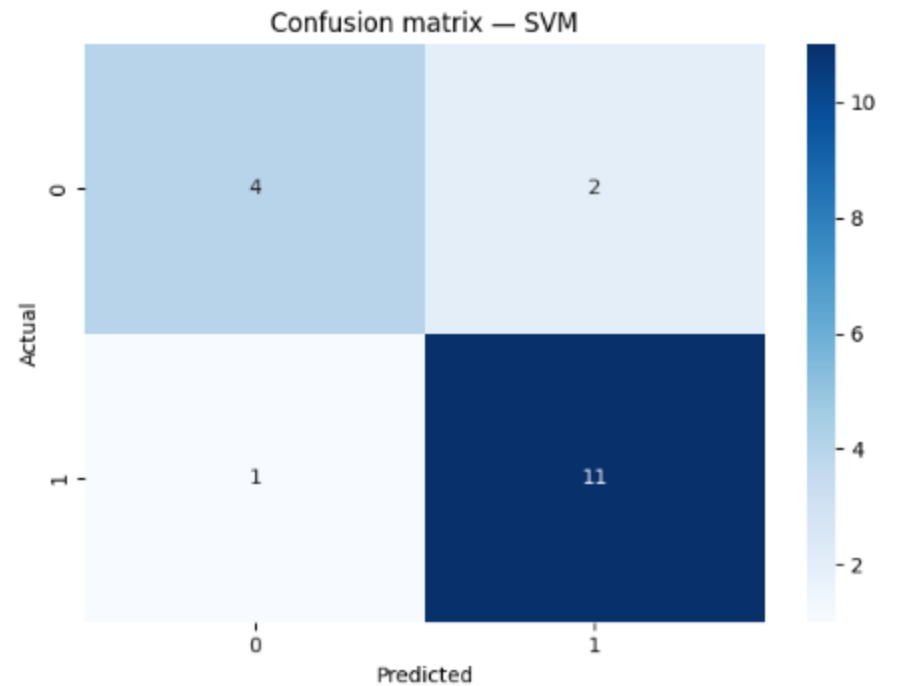
Results & Model Selection

Model	Accuracy	Precision	Recall	F1
SVM	83.3%	84.6%	91.7%	0.880
Logic Regression	77.8%	83.3%	83.3%	0.833
Decision Tree	72.2%	76.9%	83.3%	0.800

	Model	Accuracy	Precision	Recall	F1
1	SVM	0.833	0.846	0.917	0.880
0	Logistic Regression	0.778	0.833	0.833	0.833
2	Decision Tree	0.722	0.769	0.833	0.800

Best model by test F1: SVM

	precision	recall	f1-score	support
0	0.80	0.67	0.73	6
1	0.85	0.92	0.88	12
accuracy			0.83	18
macro avg	0.82	0.79	0.80	18
weighted avg	0.83	0.83	0.83	18



Conclusion & Recommendations

- Historical attributes may provide useful baseline signals.
- Validate on later launches and inspect class imbalance.
- Improve with tuning, time-based validation, more records, and monitoring.
- Treat predictions as decision support, not certainty